Research Note — *J. Geomag. Geoelectr.*, **38**, 279–284, 1986

Two Different Paleomagnetic Directions from the Izumi Group in Shikoku, Southwest Japan

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(Received July 3, 1985; Revised December 27, 1985)

1. Introduction

A large number of paleomagnetic studies have been carried out for Southwest Japan and substantiated its rapid clockwise rotation in Early Miocene (e.g., OTOFUJI and MATSUDA, 1983). However most of the previous paleomagnetic data are from Middle to Late Tertiary in age and the number of reliable pre-Tertiary data are not so sufficient to draw the continuous apparent polar wandering path of Southwest Japan. We report in this brief article the results of the paleomagnetic reconnaissance study carried out on Upper Cretaceous sedimentary rocks of the Izumi group in Shikoku, Southwest Japan (Fig. 1). The aim of the study was first to confirm the potentiality of the thick sequences of the Izumi group as a target for paleomagnetic study to obtain reliable remanence directions from representative Mesozoic sedimentary sequences in Southwest Japan. Secondly it was to examine paleomangetically, if any, the tectonic influence of the Median Tectonic Line (MTL) in the geological past because the Izumi group is clearly in fault contact to the south by the MTL with the Sanbagawa metamorphic belt yielding a clear zonal distribution as one of the major geologic belts in the outer zone of Southwest Japan. For these purposes we conducted combined geologic, paleontologic and paleomagnetic studies on a specific region in the western part of the Asan Mountains, Tokushima prefecture, where there are good representative rock facies of the Izumi group, outcropping along the north of the Yoshino river (Fig. 1).

2. Geologic Setting

The Izumi belt is a narrow zone of the Upper Cretaceous sedimentary sequences extending continuously, approximately 10 km wide, from the northwestern Shikoku to the western Kii Peninsula (KANMERA and HASHIMOTO, 1980). Structurally, the belt forms a synclinal fold with the northern limb dipping southeastward at 30° to 50° and the southern limb dipping northeastward, representing an inclined fold axis plunging about 30° to 50° to the east. To the north, the Izumi belt unconformably overlies the Ryoke metamorphic belt and, to the south, is separated from the Sanbagawa metamorphic belt by the MTL. The Izumi group is comprised mainly of

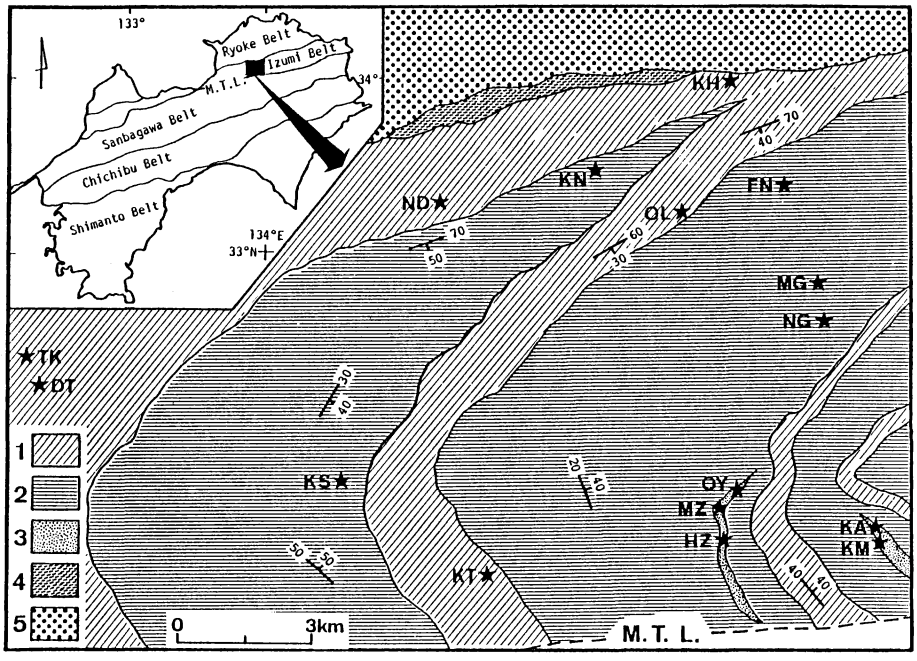


Fig. 1. Generalized geologic map of the Izumi belt in the western Asan Mountains in Shikoku, Southwest Japan, showing sampling localities for paleomagnetic studies. 1, mudstone; 2, alternations of sandstone and shale; 3, acidic tuff and tuffaceous siltstone; 4, basal conglomerate; 5, Ryoke metamorphic belt.

alternations of sandstone, shale and mudstone intercalated frequently with thin layers of acidic tuff. The age of the strata has been assigned generally to Campanian to Maastrichtian (Upper Cretaceous) by the occurrence of some ammonites from the Izumi and Asan Mountains (e.g., MATSUMOTO *et al.*, 1980), although a complete biostratigraphic correlation throughout the Izumi group has never been accomplished yet. A radiolarian age analysis carried out in this study yielded a few species of radiolaria that can be assigned to Upper Campanian (M. Okamura, 1985 personal communication). For corrections of bedding tilt necessary to the in-situ remanent magnetization vectors, we conducted an extensive geologic mapping of the studied area to determine the direction of the plunging fold axis as precisely as possible. As the result, an optimum fold axis plunging 30° to the east was estimated by the measurements of bedding pole directions (directions normal to bedding planes) from 76 localities covering almost the whole of the region.

3. Results and Discussion

A total of 16 sites as shown in Fig. 1 were visited for paleomagnetic sampling and,

in each site, one to five oriented blocks were collected by hand sampling using a magnetic compass. The rock types are mudstone (9 sites), tuffaceous siltstone (5 sites) and sandstone (2 sites). The sampling localities were selected so as to collect samples from both of the northern and southern synclinal limbs. Thus, all the sites are structurally divided into two groups, that is, the northern nine sites strike N45° to 80° E and dip 30° to 50° S while the southern seven sites strike N20° to 90° W and dip 20° to 70° N except one locality (site OY) showing an overturned bedding. Although these in-situ bedding orientations are essential to the correction of bedding tilt to the in-situ paleomagnetic directions, the conventional way by rotating the strata to horizontal about the horizontal strike of the bedding is not satisfactory in the present case because, as stated before, the folding in the studied area clearly took place about 30° plunging to the east. Therefore, the tilt correction carried out here consists of successive two procedures as following; first, rotate all the directions of the observed bedding poles and the corresponding in-situ magnetization vectors through 30° (the dip angle of inclined fold axis in the studied area) and, secondly, rotate the respective paleomagnetic direction through the angle of dip about the strike of the reoriented bedding at each site as in the same way as the conventional tilt correction.

Remanent magnetizations of 193 specimens prepared from the 16 sites were measured with a SCT cryogenic magnetometer at the Department of Geology and Mineralogy, Kyoto University. Laboratory tests for stability were carried out for several pilot specimens from each site by the progressive alternating field (a.f.) demagnetization with peak intensities from 50 Oersted (Oe) up to 1000 Oe. Since the a.f. treatment demonstrated that each pilot specimen, irrespective of the rock types, has a single stable magnetization component with the median destructive field intensity of 150 to 250 Oe, all the remaining specimens were subjected to the three steps of a.f. demagnetization of 150, 200 and 250 Oe, from which the optimum demagnetization step was defined as that producing the minimum dispersion for each site. The magnitude of natural remanent magnetization thus measured ranges from 1×10^{-7} emu/cc to 4×10^{-6} emu/cc where sandstones from the two sites are relatively more strongly magnetized about 3×10^{-6} to 4×10^{-6} emu/cc.

As a supplemental test to enhance the a.f. demagnetization, a field test was applied to all the 16 site mean directions. Table 1 summarizes the paleomagnetic results with both the mean directions before and after the tilt corrections applied following the unfolding procedures as described above, each being also illustrated in Figs. 2(a) and (b), respectively. The results show a clear difference in grouping of the mean directions that the uncorrected directions scatter generally from 10° to 80° in declination and 50° to 80° in inclination with a few extremely deflectd directions, whereas the corrected directions do not show such divergence but seem to be divisible into two groups, each corresponding to those from the northern and southern sites. That is, the nine mean directions of the northern sites cluster around 90° to 120° in declination and, in contrast, the others of the southern sites are deflected more northerly with declinations from −16° to 40°, although no significant difference in inclination can be found between the two groups. These qualitative interpretaions can be reinforced by the conventional statistical analysis resulting in that the precision

Table 1. Paleomagnetic data from the Izumi group in Shikoku.

Site	Rock type	N	I	D	I_c	D_c	k	α_{95}
KH	MS	12	67.3	60.9	47.9	113.8	105.0	4.3
OL	MS	11	57.0	65.9	38.5	93.0	379.7	2.3
KN	MS	5	68.6	75.9	51.7	115.5	490.1	3.5
ND	MS	18	72.1	46.2	52.6	101.1	49.5	5.0
DT	MS	16	53.8	54.8	48.0	96.7	181.4	2.7
TK	MS	12	52.2	73.5	40.0	101.3	113.2	4.1
MG	MS	11	71.1	2.7	58.0	111.2	311.8	2.6
NG	SS	12	52.8	48.1	43.1	84.9	104.5	4.3
FN	SS	18	59.5	27.9	55.9	88.3	211.7	2.4
KT	MS	16	82.2	325.1	67.2	27.4	344.1	2.0
KS	MS	16	73.9	49.1	41.1	35.0	255.2	2.3
MZ	TS	7	70.7	8.9	52.1	27.0	98.0	6.1
OY	TS	12	-50.5	337.7	44.5	337.5	78.4	4.9
KA	TS	10	66.4	46.4	49.5	33.8	404.3	2.4
HZ	TS	5	71.7	17.1	53.7	38.0	23.3	16.2
KM	TS	12	65.6	42.3	40.6	26.2	281.3	2.6

Rock type; MS=mudstone, SS=sandstone, TS=tuffaceous siltstone. N: number of specimens. I, D: in-situ inclination and declination. I_c, D_c: inclination and declination after correction for bedding tilt. k, α_{95}: Fisher's precision parameter and semi-angle of 95% confidence cone.

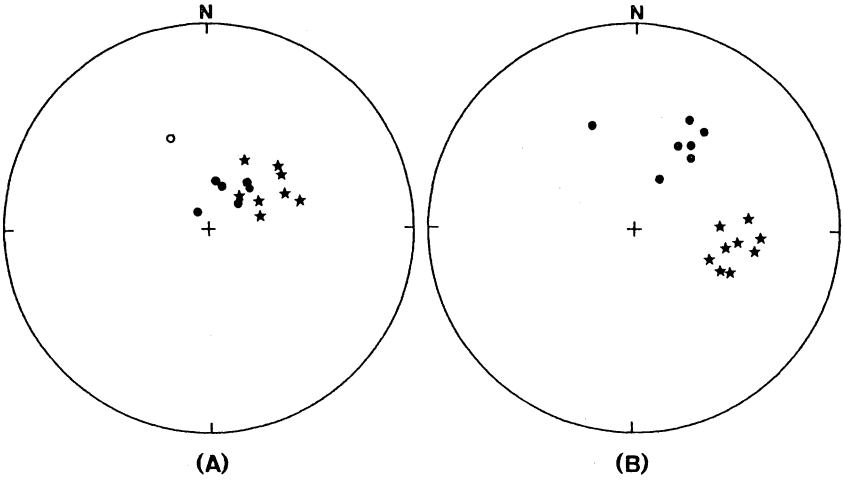


Fig. 2. Site mean directions before (a) and after (b) the corrections for bedding tilt. Circles and triangles are those from the northern and southern areas, respectively. Equal-area projection on the lower (solid) and upper (open) hemisphere.

parameters of the in-situ directions (k_b) are 40.0 and 3.7 for the northern and southern sites, respectively, while those after the tilt corrections (k_a) are raised to 66.6 and 27.6, suggesting better grouping for both groups. The ratio k_a/k_b for each was subjected to the statistical test using F-ratio tables (WATSON, 1956; MCELHINNY, 1964) resulting in that the convergence in directions of the southern and northern sites are significant at the confidence level of 99% and 80%, respectively. Although the northern sites show the lower confidence level or a smaller k_a/k_b than the southern ones, it seems that the difference in declination between the two groups may indicate the true relative tectonic rotation around the vertical axis. Alternatively, however, comparison of Figs. 2(a) and (b) may lead to another interpretation that the directions of magnetization of the entire region are secondary magnetization acquired after the folding of the beds because the in-situ directions except one (site OY) seem to show as a whole a better grouping than the corrected directions. Applying the statistics of MCFADDEN and JONES (1981) to the present case, it can be concluded significantly also that the in-situ directions can be regarded as a whole as forming one population while the corrected directions do not, that is to say that the in-situ directions may be almost secondary overprint.

Although two alternative interpretations are possible at the present stage from the statistical viewpoints as stated above, let us append some tectonic discussions to point out a possible geologic consequence of the movement of MTL to the formation of the Izumi belt. If we assume the mean direction from the northern sites in Fig. 2(b) to be the coeval paleomagnetic direction of Southwest Japan, then the direction from the southern sites with less eastward deflectd declination suggests the counter clockwise rotation about the vertical axis which may be attributable to the horizontal drag due to the strike-slip movement of MTL. Tectonically, taking the pre-Tertiary movement of MTL inferred by ICHIKAWA (1980) into consideration, this rotation may mean the left-lateral displacement of MTL which may have caused the plastic deformation near the fault zone prior to the folding of the Izumi group of this area.

The author is grateful to Dr. M. Torii (Kyoto University) for allowing the use of cryogenic magnetometer and other equipments in his laboratory during the course of this study. Mr. M. Nozoe is also acknowledged for his assistance in the field and laboratory works. This study was supported in part by the grant-in-aid from the Ministry of Education (No. 59740187).

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